AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

APRIL/ MAY 2017

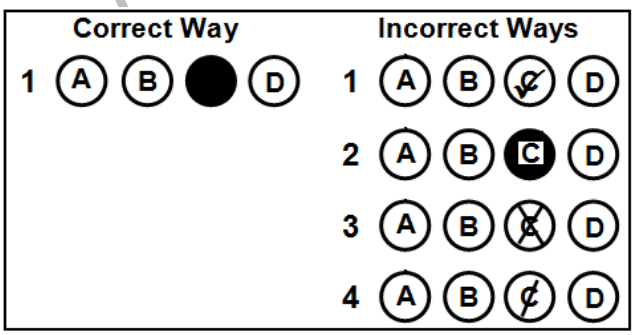
Biology Paper I

Time: 35 minutes Marks: 25

AKU-EB May Examination 2017

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

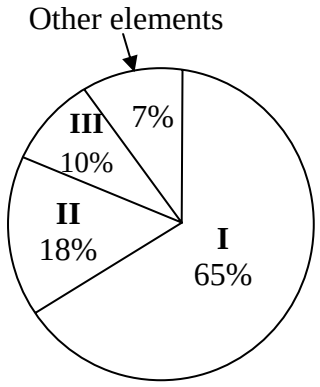
5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

1. A group of students investigated the chemical composition of soil from different cities of Pakistan. The students gathered data about the presence of relative percentage of nutrients in the soil and presented the data in the form of tables and graphs.

   The branch of biology involved in the presentation of data in the given scenario is

   A. biochemistry.
   B. biogeography.
   C. bioeconomics.
   D. biomathematics.

2. The given pie chart shows the distribution of bio-elements in the human body.



   The elements represented by **I**, **II** and **III** are

	I	II	III
A	Oxygen	Carbon	Hydrogen
B	Carbon	Oxygen	Hydrogen
C	Hydrogen	Carbon	Oxygen
D	Carbon	Hydrogen	Oxygen

3. Four different levels of biological organisation are represented in the given diagrams.



I **II** **III** **IV**

   The CORRECT sequence in the descending order of biological organisation is

   A. III, II, I, IV
   B. IV, I, II, III
   C. III, I, II, IV
   D. IV, II, I, III

AKU-EB May Examination 2017

4. According to the scientific method of solving a biological problem, which of the following CANNOT be opposed or rejected?

   A. Mass-flow hypothesis
   B. Mendel's laws of inheritance
   C. Darwin's theory of evolution
   D. A student's daily observations

5. A scientist is conducting an experiment to test the hypothesis that a species of plant will produce more fruits when exposed to blue wavelength of light at night. The scientist takes 20 experimental plants that are exposed to blue light at night. The control group should

   A. be exposed to blue light all the time.
   B. not be exposed to blue light at night.
   C. not be exposed to blue light during the day.
   D. be exposed to a different wavelength of light, like red.

6. Biodiversity in an ecosystem is advantageous because it

   A. decreases rainfall in plains.
   B. increases productivity of food.
   C. increases competition for food.
   D. decreases decomposition of organic matter.

7. Biological classification of pea is given in sequence.

   - Plantae
   - Magnoliophyta
   - Magnoliopsida
   - Fabales
   - Fabaceae
   - *Pisum*

   The order to which pea belongs is

   A. Fabales.
   B. Fabaceae.
   C. Magnoliopsida.
   D. Magnoliophyta.

8. A biology student observes a plant with the following features.

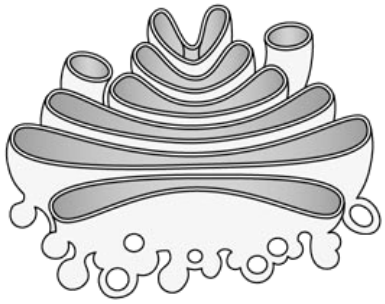
   - It contains rhizoids.
   - It reproduces by spores.
   - It contains vascular tissues.

   The plant is identified as a

   A. fern.
   B. moss.
   C. dicot.
   D. monocot.

AKU-EB May Examination 2017

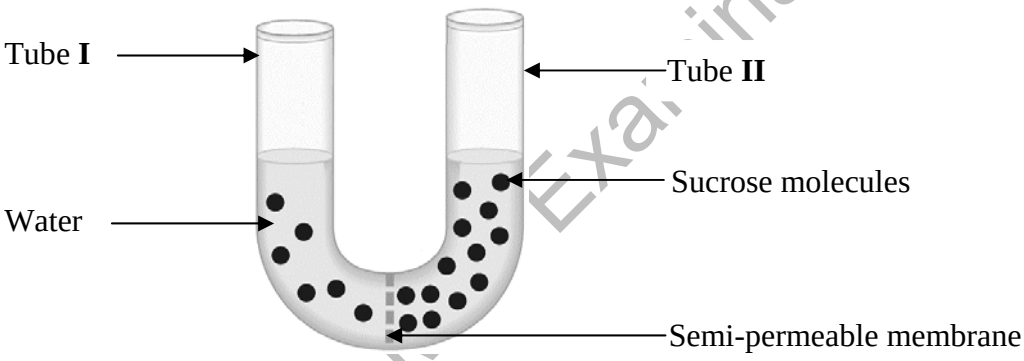
9.



Which organelle is shown in the given diagram?

A. Centrioles
B. Golgi bodies
C. Rough endoplasmic reticulum
D. Smooth endoplasmic reticulum

10. The given apparatus shows a setup to study the movement of material across a semi-permeable membrane.



After three hours, the

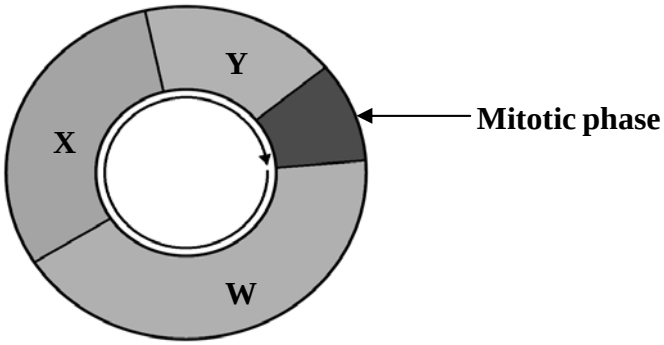
A. level of water in tube **I** will rise.
B. level of water in tube **II** will rise.
C. number of sucrose molecules in tube **I** will increase.
D. number of sucrose molecules in tube **II** will increase.

11. The CORRECT examples of simple and compound tissues in plants are

	Simple Tissue	**Compound Tissue**
A	Collenchyma	Xylem
B	Phloem	Epidermis
C	Epidermis	Collenchyma
D	Xylem	Phloem

AKU-EB Examination 2017

12. The given diagram shows the cell cycle in an animal.



If the stage labelled as **X** does NOT take place, then the

A. synthesis of DNA would stop.
B. supply of proteins would stop.
C. spindle fibres would not be formed.
D. cell organelles would not be doubled.

13. Which of the given diagrams represent **prophase II** of meiosis?



A B C D

14. When a human germ cell undergoes the process of meiosis, in the end, each cell contains 23

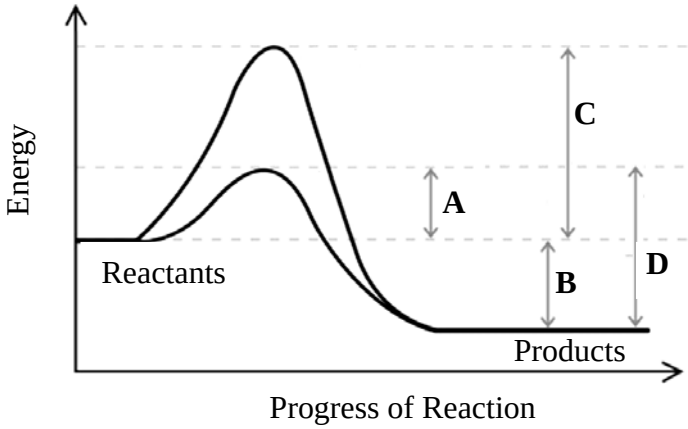
A. homologous chromosomes with one chromatid each.
B. homologous chromosomes with two chromatids each.
C. non-homologous chromosomes with one chromatid each.
D. non-homologous chromosomes with two chromatids each.

15. Which of the following represents the characteristics of necrosis?

	Cell Shrinkage	Chromatin Condensation	Lysosomal Leakage
A	×	×	✓
B	×	✓	✓
C	✓	✓	×
D	✓	×	×

AKUEB Matric Examination 2017

16. Which of the following shows the activation energy of an enzyme catalysed reaction?



17. The chemical reaction that requires extra-cellular enzymes is the

A. breakdown of fats.
B. oxidation of glucose.
C. synthesis of proteins.
D. reduction of carbon dioxide.

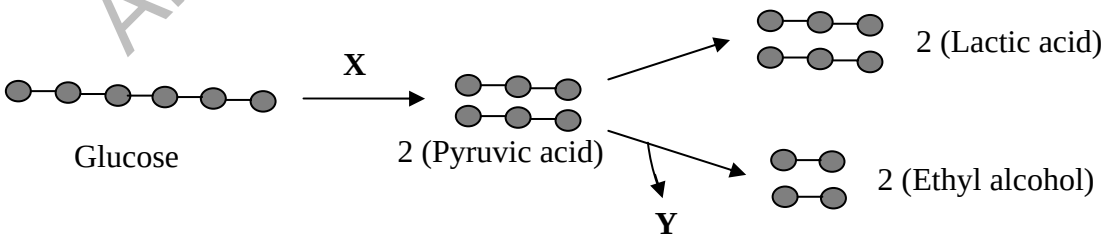
18. When a molecule of adenosine diphosphate reacts with water, it gives

A. adenosine triphosphate and hydrogen gas.
B. adenosine monophosphate and hydrogen gas.
C. adenosine triphosphate and inorganic phosphate.
D. adenosine monophosphate and inorganic phosphate.

19. How many water molecule(s) are involved in light dependent reactions to produce TWO oxygen molecules by the end of photosynthesis?

A. 1
B. 2
C. 3
D. 4

20. The given diagram shows some events of anaerobic respiration.



Which process and by-product formation occurs at **X** and **Y** respectively?

	X	Y
A	Glycolysis	Carbon dioxide
B	Glycolysis	Adenosine triphosphate
C	Alcoholic fermentation	Carbon dioxide
D	Alcoholic fermentation	Adenosine triphosphate

AKU-EB May Examination 2017

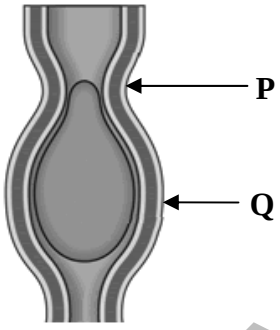
21. The given table shows estimated energy requirements (in kilocalories) according to age, gender and activity.

Gender	Age (years)	Activity Level		
		Sedentary	Moderately Active	Active
Male	4-8 (child)	1,400 (kcal)	1,400-1,600 (kcal)	1,600-2,000 (kcal)
	31-50 (adult)	2,200 (kcal)	2,400-2,600 (kcal)	2,800-3,000 (kcal)

The information in the given table implies that

A. the energy requirement of all males is the same.
B. adult males need to do more work than children.
C. children need to burn more calories than adult males.
D. adult males should take more carbohydrates than children.

22. The given diagram shows peristalsis in oesophagus.



The condition of longitudinal muscles at points **P** and **Q** would be

	P	Q
A	Contract	Relax
B	Relax	Contract
C	Contract	Contract
D	Relax	Relax

AKU-EB May Examination 2017

23. Diagram **I** represents a plant in wilted state while diagram **II** represents the same plant in the normal state.



Diagram I



Diagram II

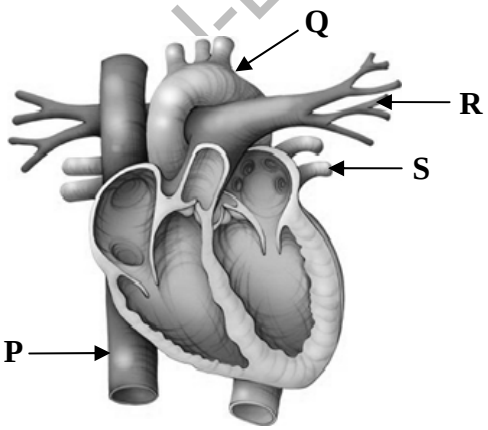
Which of the following are required for the plant shown in diagram **I** to regain condition shown in diagram **II**?

	Rate of Absorption	**Rate of Transpiration**
A	Increase	Decrease
B	Decrease	Increase
C	Decrease	Decrease
D	Increase	Increase

24. Which of the following is CORRECT about thrombocytes?

	Nucleus	**Pigment**	**Life span**
A	Absent	Absent	7-8 days
B	Present	Absent	120 days
C	Absent	Present	7-8 days
D	Present	Present	120 days

25. The given diagram represents the human heart.



The blood vessel that receives the oxygenated blood from lungs is

A. P
B. Q
C. R
D. S

END OF PAPER

I-EB May Examination 2017